

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines which allows them to think and act like humans. It involves creating algorithms and systems that can perform tasks that require human abilities such as visual perception, speech recognition, decision-making and language translation.

It allows systems to learn from data, recognise patterns, and make decisions to solve complex problems.

Foundations of AI

AI is built upon several major disciplines:

Foundation	Contribution to AI
Philosophy	Logic, reasoning, knowledge, and principles of intelligent behavior
Mathematics	Probability, statistics, logic, optimization, and algorithms
Economics	Decision-making, utility, and optimization
Neuroscience	Understanding how the human brain learns and processes information
Psychology	Human learning, perception, memory, and problem-solving
Computer Science	Algorithms, programming, data structures, and computation
Control Theory	Feedback, stability, and autonomous decision-making
Linguistics	Understanding and processing human language

Limitations of Artificial Intelligence

Although **Artificial Intelligence (AI)** can perform many complex tasks, it still has several limitations. AI does not possess human intelligence in the complete sense and depends heavily on **data**, **algorithms**, **computing resources**, and **human-designed objectives**.

1. Lack of Common Sense

AI can process information and identify patterns, but it may not understand basic real-world situations like humans.

Example: An AI may understand the words in a sentence but misunderstand the actual situation or intention.

2. Dependence on Data

AI systems, especially Machine Learning models, require large amounts of high-quality data.

- Poor data → Poor predictions
- Biased data → Biased results
- Insufficient data → Limited performance

3. Lack of True Understanding

AI can generate convincing answers without necessarily **understanding** the subject in the human sense.

For example, an AI language model can explain emotions but does not experience emotions itself.

4. Lack of Emotions and Empathy

AI does not genuinely feel emotions such as:

- Happiness ,Fear ,Love ,Sympathy ,Anger

5. High Cost

Developing and operating advanced AI systems can require:

- Powerful GPUs
- Large storage
- High electricity consumption
- Large datasets
- Skilled AI engineers

6. Security and Privacy Risks

AI systems can introduce risks such as:

- Data privacy violations
- Deepfakes
- Automated misinformation
- Adversarial attacks
- Misuse of personal data

AI ETHICS

Artificial Intelligence is now used in many important areas across different industries. As these systems grow, ethical rules are needed to prevent misuse, protect user data and avoid unfair decisions. AI ethics provides guidelines to keep AI reliable, responsible and safe.



Principle	Meaning	Example
1. Fairness	AI should treat people fairly and avoid discrimination.	A hiring AI should not reject candidates based on gender or background.
2. Transparency	Users should know when and how AI is being used.	A company should disclose that a decision was made using AI.

3. Accountability	Someone must be responsible for AI's decisions and outcomes.	Developers and organizations should take responsibility for harmful AI decisions.
4. Privacy	AI should protect users' personal and sensitive information.	Facial-recognition systems should handle personal data carefully.
5. Safety	AI systems should operate reliably and avoid causing harm.	An autonomous vehicle should prioritize human safety.
6. Explainability	AI decisions should be understandable, especially in high-risk applications.	A bank should be able to explain why an AI rejected a loan application.

Future of AI – One-Line Points

1. **Advanced AI:** AI will become more intelligent and capable of solving complex problems.
2. **Generative AI:** AI will create text, images, videos, music, and code.
3. **AI Agents:** AI will independently plan and perform multi-step tasks.
4. **Healthcare:** AI will help in disease detection, diagnosis, and drug discovery.
5. **Robotics:** AI-powered robots will perform more complex physical tasks.
6. **Autonomous Vehicles:** AI will improve self-driving cars and transportation systems.
7. **Education:** AI will provide personalized learning and intelligent tutoring.
8. **Business:** AI will automate tasks and improve business decision-making.
9. **Multimodal AI:** AI will understand and combine text, images, audio, and video.
10. **Scientific Research:** AI will accelerate discoveries in medicine, science, and engineering.
11. **Human-AI Collaboration:** Humans and AI will work together to improve productivity and decision-making.

AI Components

AI consists of several components that work together to make a system **intelligent and capable of solving problems**.

1. **Knowledge Representation:** Represents and stores facts, rules, and information so that the AI system can use them.
Example: Storing information about diseases and their symptoms.
2. **Reasoning:** Uses stored knowledge and logical rules to derive conclusions.
Example: If a patient has certain symptoms, AI can reason about possible diseases.
3. **Learning:** Allows AI to learn from data and improve its performance without being explicitly programmed for every situation.
Example: A spam filter learns to identify spam emails from previous examples.
4. **Problem Solving:** Finds a suitable solution to a problem by exploring possible alternatives.
Example: Finding the shortest route between two locations.
5. **Planning:** Determines a sequence of actions needed to achieve a particular goal.
Example: A robot plans the steps required to move an object from one place to another.
6. **Perception:** Enables AI to understand information received from the environment through sensors, cameras, microphones, etc.
Example: A self-driving car detects pedestrians and traffic signs using cameras.
7. **Natural Language Processing (NLP):** Enables computers to understand, process, and generate human language.
Example: Chatbots understanding and responding to user questions.

8. **Decision Making:** Selects the most appropriate action from available choices based on goals and information.
Example: A recommendation system deciding which movie to suggest.
9. **Action:** Allows an AI system to perform the selected action in the physical or digital environment.
Example: A robot moving its arm after deciding where an object is located.

Types Of Ai Agent

1. Simple reflex agents act only on the current perception of the environment using predefined condition–action rules. They do not rely on past experiences or predict future outcomes and respond directly using simple “if–then” logic.

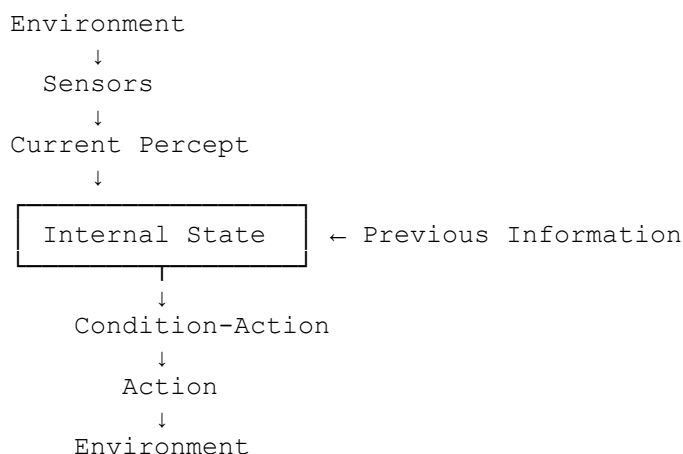
- Operate without memory of past states or stored information
- Do not maintain any internal model of the environment
- Work best in fully observable and predictable environments

For Example, Traffic light control systems that change signals based on fixed timing

2. Model-Based Reflex Agent

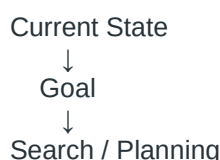
A **Model-Based Reflex Agent** maintains an **internal state or model of the environment**. It uses the current percept along with information about previous situations to make decisions.

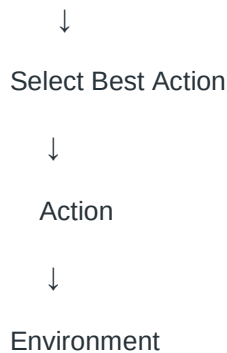
Diagram



Goal-based agents choose their actions by focusing on a specific objective and evaluating how different choices can help achieve it. Instead of reacting only to the current situation, they plan ahead and consider possible future outcomes.

- Use search and planning techniques to find suitable action paths
- Judge actions based on how well they move toward the goal
- Consider future consequences before making decisions
- Explore multiple possible routes to reach the desired outcome

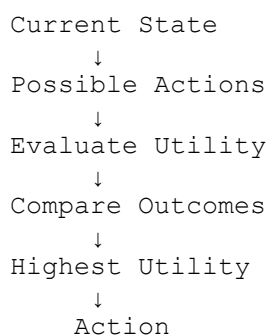




4. Utility-Based Agent :A **Utility-Based Agent** not only tries to achieve a goal but also evaluates **how good or useful each possible outcome is**.

It uses a **utility function** to select the action that provides the highest expected utility.

Diagram

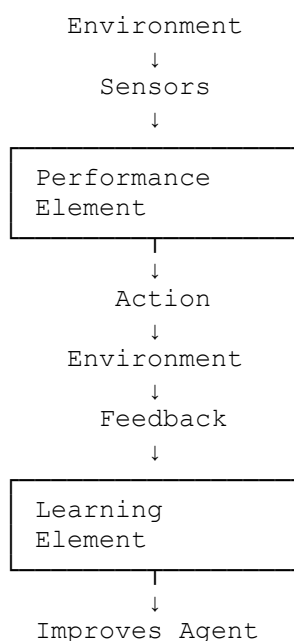


5. Learning Agent

A **Learning Agent** can **learn from experience and feedback** and improve its performance over time.

Unlike fixed-rule agents, a learning agent can modify its behavior based on what it learns.

Diagram



Unit 2

Greedy Best first search algorithm

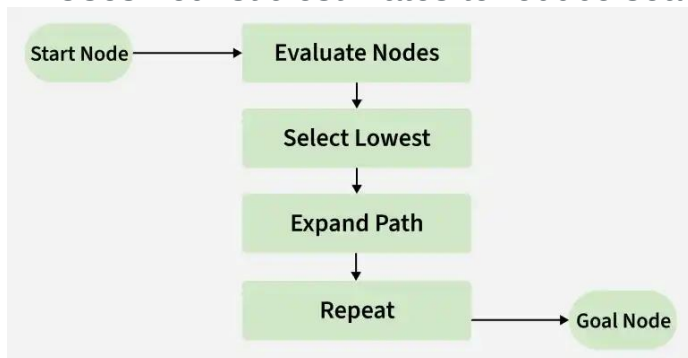
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- Greedy Best-First Search is an AI search algorithm used to find a path from a starting node to a goal node by selecting the path that appears most promising at each step. It uses heuristic values to guide the search toward the goal efficiently.
 - Prioritizes faster exploration over guaranteed optimal solutions
 - Expands the node with the lowest heuristic value
 - Reduces search effort by focusing on promising paths
 - Commonly applied in navigation systems and pathfinding problems

Working

Greedy Best-First Search works by selecting the node that appears closest to the goal using heuristic values and continues expanding the most promising path until the destination is reached.

- Starts from the initial node and evaluates neighboring nodes using heuristic values
- Selects the node with the lowest heuristic value for expansion
- Expands the most promising path instead of exploring all possible paths
- Repeats the process until the goal node is reached
- Uses heuristic estimates to reduce search time and guide traversal efficiently



Advantages

- Easy to implement due to its simple heuristic-based approach
- Finds solutions quickly by expanding the most promising nodes first
- Requires comparatively less memory than many other search algorithms

Limitations

- Does not guarantee the shortest or most optimal solution as it only focuses on the most promising path
- Can get trapped in local optima, leading to suboptimal path selection
- Requires a suitable heuristic function, which may increase algorithm complexity
- May fail in highly complex or cyclic search spaces without additional handling mechanisms

Applications

- **Pathfinding:** Used to find efficient paths in video games, robotics, and navigation systems
- **Machine Learning:** Helps explore promising solutions within large search spaces
- **Optimization:** Assists in selecting parameter values for desired outcomes
- **Game AI:** Evaluates possible moves to choose favorable actions
- **Natural Language Processing:** Applied in tasks like language translation and speech recognition
- **Image Processing:** Helps segment images into meaningful regions

A* Search Algorithm

1. Definition

A* (A-star) Search is an **informed (heuristic) search algorithm** used to find the **least-cost path from a start state to a goal state**.

It combines:

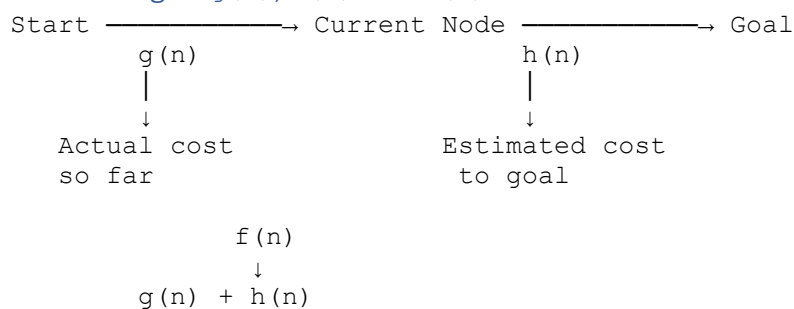
- **Actual cost** already travelled from the start $\rightarrow g(n)$
- **Estimated cost** remaining to the goal $\rightarrow h(n)$

The evaluation function is:

$$f(n) = g(n) + h(n)$$

The algorithm always selects the node with the **lowest $f(n)$ value** for expansion.

2. Meaning of $g(n)$, $h(n)$ and $f(n)$



$g(n)$ – Actual Cost

The actual cost of reaching node n from the starting node.

$h(n)$ – Heuristic Cost

The estimated cost of reaching the goal from node n .

$f(n)$ – Total Estimated Cost

The estimated total cost of a solution through node n .

Formula:

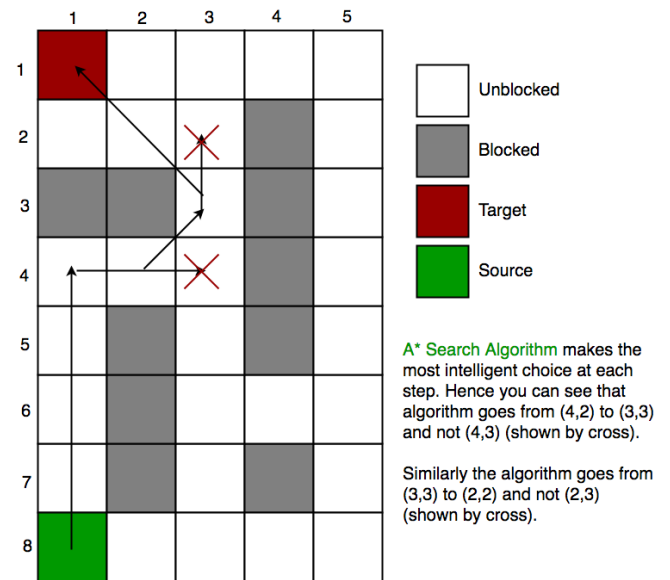
$$f(n) = g(n) + h(n)$$

3. Working of A* Search

A* maintains a set of candidate nodes, usually called the **OPEN list/frontier**, and repeatedly selects the node with the smallest $f(n)$.

Steps

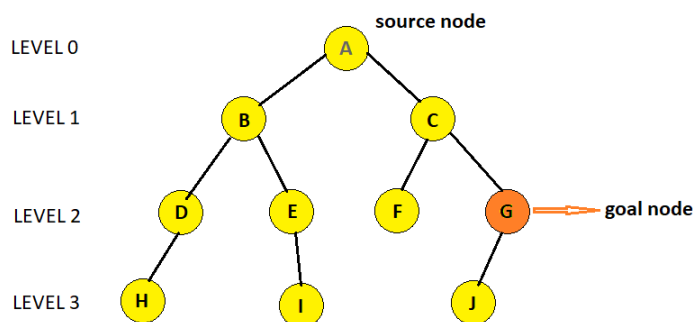
1. Start from the initial node.
2. Calculate $g(n)$, $h(n)$, and $f(n)$.
3. Select the node with the **smallest $f(n)$** .
4. Expand that node and generate its successors.
5. Calculate their g , h , and f values.
6. Add promising successors to the frontier.
7. Continue until the goal is selected.
8. Return the path to the goal.



Iterative Deepening Search(IDS) or Iterative Deepening Depth First Search(IDDFS)

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- There are two common ways to traverse a graph, **BFS** and **DFS**. Considering a Tree (or Graph) of huge height and width, both BFS and DFS are not very efficient due to following reasons.
 - DFS** first traverses nodes going through one adjacent of root, then next adjacent. The problem with this approach is, if there is a node close to root, but not in first few subtrees explored by DFS, then DFS reaches that node very late. Also, DFS may not find shortest path to a node (in terms of number of edges).
 - BFS** goes level by level, but requires more space. The space required by DFS is $O(d)$ where d is depth of tree, but space required by BFS is $O(n)$ where n is number of nodes in tree (Why? Note that the last level of tree can have around $n/2$ nodes and second last level $n/4$ nodes and in BFS we need to have every level one by one in queue).
- IDDFS** combines depth-first search's space-efficiency and breadth-first search's fast search (for nodes closer to root).



IDDFS with max depth-limit = 3
 Note that iteration terminates at depth-limit=2
Iteration 0: A
Iteration 1: A->B->C
Iteration 2: A->B->D->E->C->F->G